MIL-S-5676A

4 April 1950
Superseding
MIL-S-5676
27 February 1950

MILITARY SPECIFICATION

SPLICING; CABLE TERMINAL, PROCESS FOR, AIRCRAFT

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE AND CLASSIFICATION

1.1 This specification is designed to present detailed procedure to be followed in splicing cable terminals. The types of cable to which this specification applies are listed in Section 2 and table I.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 The following specifications and drawings of the issue in effect on date of invitation for bids shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Federal

QQ-S-571	Solder; Soft (Tin, Tin-Lead, and Lead-Silver)
V-T-276	Thread; Cotton

Military

MIL-C-1511	Cable; Steel (Carbon), Flexible Preformed
MIL-C-5424	Cable; Steel (Corrosion-Resisting), Flexible Preformed (For Aeronautical Use)

Air Force-Navy Aeronautical

AN-C-52	Compound; Exterior-Surface, Hard Film, Corrosion-Preventive
AN-C-76	Cable; Single Strand Nonflexible Carbon Steel (Preformed)
AN-W-22	Wire; Zinc Coated, Soft Steel

2.1.2 Drawings.-

Air Force-Navy Aeronautical Standard Drawing

AN100	Thimble - Wire Cable

(Copies of this publication and copies of other publications referenced herein or required for Government procurement, and the Index of Military Aeronautical (AN or MIL) Standards, may be obtained upon application to the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio; or the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania.)

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3. MATERIAL AND EQUIPMENT

3.1 Materials.-

3.1.1 Thimbles.- Thimbles shall conform to the requirements of Drawing AN100.

3.1.2 Serving Cord.- Serving cord shall conform to Specification V-T-276, type 111A, 12-ply.

3.1.3 Preservative.- The preservative shall conform to the requirements of Specification AN-C-52, type 1.

3.1.4 Serving Wire.- Serving or wrapping wire shall be in accordance with Specification AN-W-22.

3.1.5 Solder.- Solder shall conform to Specification QQ-S-571, class A.

3.1.6 Flux.- Solder flux shall be a compound of stearic acid (free from mineral acids) and rosin of the following proportions:

Stearic Acid	25 to 50%
Rosin	50 to 75%

3.2 Equipment.- The following tools (see figure 14) are recommended for use in the preparation of cable terminal splices.

3.2.1 Marlin Spike.- This is a small pointed tool of oval cross-section with a screw driver handle.

3.2.2 Side Cutting Pliers.- The side cutting pliers are required for clipping off the strands after splicing.

3.2.3 Splicing Clamp.- A splicing clamp is useful for holding the cable in place in the thimble and in positioning the terminal during splicing operations.

3.2.4 Mallet.- A rawhide or hardwood mallet is needed for hammering the splice after finishing and before serving, in order to produce a neat looking splice.

4. PREPARATION AND PROCEDURE

4.1 Cutting of Cable.-

4.1.1 Flexible Cable.- The cable shall be cut to length by mechanical means only, such as a fine tooth hacksaw. The use of oxy-acetylene torches in any manner is not permitted.

4.1.2 Non-Flexible Cable.- Before cutting the cable the wires must be soldered or welded together to prevent slipping. The preferable process is to thoroughly tin and solder the cable for 2 to 3 inches by placing in a solder trough as shown in figure 15 and finishing smooth with a soldering tool. The cable may be cut diagonally to conform to the required taper finish shown in figure 16. The cable shall be cut to length by mechanical means, such as a fine tooth hacksaw, except as allowed in applicable sub-paragraphs under "Procedure."

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4.2 Procedure.-

4.2.1 Operations for Flexible Cable.- The procedure for the preparation of terminal splices for flexible cable is described in the following paragraphs. The splice shall have five full tucks. For 1/16-inch 7 x 7 flexible steel cable, the procedure described herein for nonflexible cable may be used as an optional method.

4.2.1.1 Forming of Terminals.- The cable shall be bent snugly around the proper size of thimble as specified in table I and clamped, the points of the thimble having previously been bent back at an angle of approximately 45 degrees to permit a close splice. If a regular splicing clamp is not available, the cable may be held in place on the thimble by binding with cord, as shown in the accompanying illustrations. The length of the free end of the cable should be about 8 inches for sizes up to and including 5/32-inch 7 x 19 cable. For cable of larger sizes the length should be slightly greater. The standing wire should be clamped in a vise with the free end to the left of the standing wire and away from the operator.

4.2.1.2 First Operation.- Choose the free strand lying directly under the thimble points and separate it from the rest of the free ends by inserting the Marlin spike under it and spiralling the spike around the free end. The cable should then appear as in figure 1.

4.2.1.3 Second Operation.- Insert the Marlin spike under the first three strands of standing cable to the right of the thimble points. Insert the loose free strand under the three standing strands, tucking from left to right. (Figure 2 illustrates the condition of the cable at this point.)

4.2.1.4 Third Operation.- Choose the second free strand to the left of the thimble points and separate it from the others as before described. This wire should be the next to the left of the first free strand tucked. Insert the Marlin spike under the first two standing strands to the right of the thimble points and tuck the loose end under these two standing strands, tucking from left to right as in figure 3.

4.2.1.5 Fourth Operation.- Choose the third free strand to the left of the thimble points and separate as before. Insert the Marlin spike under one standing strand to the right of the thimble points and tuck the loose free end under this standing strand, tucking from left to right. The cable now appears as in figure 4.

4.2.1.6 Fifth Operation.- Choose the center or core wire and separate it from the other free ends. It can be easily distinguished from the outside wires by its smooth even surface. The outside wires are waved evenly by being coiled around the center or core wire. Tuck the center free end under the last two standing strands used, and bring it out in the same opening as the second free end which was tucked. The core wire should lie between the vise and the second free end as appears in figure 5.

4.2.1.7 Sixth Operation.- Choose the free strand which lies last to the left of the thimble points and separate as before. Insert the Marlin spike from right to left under the two standing strands immediately to the left of the thimble points. Insert the loose free end from right to left under these two standing strands. This is the only place in the first round of tucks where the tuck is made from right to left. The cable should now appear as in figure 6. To complete the sixth operation, pull the free end of the last strand tucked, until the loop shown in figure 6 disappears and there is no slack in the strand.

4.2.1.8 Seventh Operation.- Separate the two remaining free ends. Insert the Marlin spike under the first standing strand to the left of the thimble points. Choose the free end which lies next in order of rotation to the first three wires tucked. This should be the fourth free strand to the left of the thimble points. Tuck this free strand under the lifted standing strand, tucking from left to right. Figure 7 illustrates the present condition of the splice.

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4.2.1.9 Eighth Operation.- With the Marlin spike, lift the second standing strand to the left of the thimble points. Tuck the remaining free strand under the lifted standing strand. Tuck from left to right. The cable should now appear as in figure 8.

4.2.1.10 Ninth Operation.- The finished condition of a splice depends considerably on the ninth operation, so it should be observed very closely. Choose the tucking strand which lies in the opening with the center strand. Grasp it firmly in a pair of fairly heavy pliers and pull sharply upon it several times away from the thimble. Then keeping the strand always taut, describe a half circle with the pliers and pull steadily back into the throat of the thimble. (The throat of the thimble is the joint of the two ends lying between the four points, previously turned back at a 45 degree angle.) Continue this operation completely around the standing cable until each of the tucking strands has been pulled down away from the thimble and then back aginst the throat. Next pull down the free center strand, but do not pull it back.

4.2.1.11 Tenth Operation.- Choose the tucking strand to the right of the center free strand and tuck it over one standing strand and under the next standing strand. Tuck from right to left. See figure 9. Continue this operation around the standing cable to the left until you have tucked all six of the tucking strands. Do not tuck the free center strand on this or any succeeding series of tucks. After tucking all six tucking strands repeat the ninth operation. This constitutes the second series of tucks and the cable should appear as in figure 10.

4.2.1.12 Eleventh Operation.- The third series of tucks is made exactly the same as the second series. (Followed by the ninth operation).

4.2.1.13 Twelfth Operation.- The six tucking strands are now split in half, and a series of tucks are made in the same manner as that described for the full strands, but using only one-half of each strand. (Followed by the ninth operation.) See figure 11.

4.2.1.14 Thirteenth Operation.- The core wire or free center strand should now be cut off as close as possible to the standing cable. The six half strands which were tucked in the twelfth operation are again halved making them 1/4 the size of a full strand. A series of tucks is now made using six 1/4 strands. (Followed by the ninth operation.) See figure 12. This completes the tucking of the strands.

4.2.1.15 Pounding the Splice.- Bend the thimble points back into position and tap them snugly against the splice. Using a hardwood anvil and a hardwood or rawhide faced mallet, start at the thimble points and drive the tucking strands from right to left just as they were tucked. Roll the splice on the anvil while pounding in order that the strands may receive an even tautness. When the splice has been made taut and hard, cut off the tucking strands as close as practicable. Pound the taper slightly to lay the sheared ends.

4.2.1.16 Serving the Splice.- Start about 1/4-inch below the end of the taper and wrap smoothly up the taper until all of the sheared ends are covered. Then take five or six loose coils over thumb or finger and insert the end of the serving cord through the loose coils from the thimble and toward the taper. Now wrap the loose coils firmly over the inserted end and pull up any slack by drawing the inserted end down toward the taper. Cut off close to the serving, and beat it lightly with the mallet to smooth it down. Two coats of shellac shall be applied to the serving to make it as nearly waterproof as possible. The finished splice should appear as in figure 13.

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4.2.2 Operations for Non-Flexible Cable.- The procedure for the manufacture of terminal splices for non-flexible cable is described in the following paragraphs. This terminal must not be used on flexible cable, except that it may be used for 1/16-inch 7 x 7 flexible cable.

4.2.2.1 Forming of Splice.- After soldering and cutting, the cable is securely bent around the proper size thimble and clamped, taking care that the cables lie close and flat, and that the taper end for finish lies on the outside. If it is necessary to trim the taper at this point in the process, it is preferable that it be done by nipping, but grinding will be permitted provided a steel guard at least 3 inches long and 1/32 inch thick is placed between the taper end and the main cable during the operation, and that the heat generated from the grinding does not melt the solder and loosen the wires.

4.2.2.2 Serving.- Serving may be done by hand or machine, but in either case each serving convolution must touch the adjoining one and be pulled tightly against the cable, with spaces for permitting a free flow of solder and inspection, as shown in figure 16. Serving wire shall be as specified in Section 3 and figure 16.

4.2.2.3 Soldering.-

4.2.2.3.1 Care must be exercised to prevent drawing of the temper of any cable wires by excessive temperature or duration of applied heat. The flux used in this soldering shall be as specified in Section 3. Salammoniac or other compound having a corrosive effect will not be permitted either as a flux or for cleaning the soldering tools.

4.2.2.3.2 Soldering shall be accomplished by immersing the terminal alternately in the flux and in the solder bath (figure 15), repeating the operation until through tinning and filling with solder under the serving wire and thimble is obtained. The temperature of the solder bath and place where the terminal is withdrawn shall not be above 450°F. A soldering iron may be used in the final operation to give a secure and good appearing terminal. Care must be taken that the solder completely fills the space under the serving wire and thimble. A slightly hollowed cast-iron block to support the splice during soldering may help in securing best results. Abrasive wheels or files for removing excess solder will not be permitted.

4.2.3 Alternate Process for Non-Flexible Cable.- As an alternate process of making terminals for non-flexible cable, the oxy-acetylene cutting method and the pre-soldering method (soldering before wrapping) are permitted, but only on the following conditions:

(a) That the process of cutting securely welds all wires together.

(b) That the annealing of the cable does not extend more than one cable diameter from the end.

(c) That no filing be permitted either before or after soldering.

(d) For protection during the operation of grinding the tapered end of the cable, a steel guard at least 3 inches in length and 1/32 inch thick shall be placed between the taper and the main cable.

(e) The heat from grinding shall not draw the temper of the cable.

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5. SAMPLING, INSPECTION, AND TEST PROCEDURES

5.1 General.- All cable terminal splices shall be subject to inspection by authorized Government Inspectors who shall be given all reasonable facilities to determine conformance with the requirements of this specification.

5.2 Sampling.- All cable terminal splices shall be subjected to the Proof Load test specified hereinafter. One half of one percent, but in no case less than one sample, of all terminal splices in each size of cable shall be selected for subjection to the Full Strength test specified herein.

5.3 Tests.-

5.3.1 Proof Load.- The splice and the entire length of the cable shall be subjected to the proof load specified in the following table. The load shall be applied gradually and continuously until the full proof load is attained, after which the proof load shall be applied for a period of not less than 3 minutes. Assemblies for cable sizes not shown in the following table shall be subjected to a proof load equal to 60 percent of the breaking load specified in the specification covering the cable involved.

Test Loads for Cable Terminal Splices

Cable Diameter Inch	Thimble		Flexible Cable				Non-Flexible Cable	
			Specification MIL-C-1511		Specification MIL-C-5424		Specification AN-C-76	
	Carbon Steel Cable	Cor. Resist. Steel Cable	Proof Load Pounds	Full Strength Pounds	Proof Load Pounds	Full Strength Pounds	Proof Load Pounds	Full Strength Pounds
1/16	AN100-3	AN100C-3	300	410	300	410	300	475
3/32	AN100-4	AN100C-4	550	780	550	780	700	1,040
1/8	AN100-4	AN100C-4	1,200	1,700	1,150	1,610	1,250	1,990
5/32	AN100-5	AN100C-5	1,700	2,380	1,550	2,210	1,900	3,040
3/16	AN100-6	AN100C-6	2,500	3,570	2,350	3,310	2,750	4,370
7/32	AN100-7	AN100C-7	3,350	4,760	3,100	4,420	3,650	5,800
1/4	AN100-8	AN100C-8	4,200	5,950	3,950	5,610	4,800	7,600
9/32	AN100-9	AN100C-9	4,800	6,800	4,800	6,800	--	--
5/16	AN100-10	AN100C-10	5,900	8,320	4,900	6,960	7,500	11,900
3/8	AN100-12	AN100C-12	8,650	12,200	7,200	10,200	10,500	16,600
7/16	AN100-14	AN100C-14	10,600	15,000	9,600	13,600	14,100	22,300
1/2	AN100-16	AN100C-16	13,700	19,400	13,700	19,400	17,100	27,100

5.3.2 Full Strength.-

5.3.2.1 Flexible Cable.- The sample splices shall be subjected to the Full Strength load specified in the above table. This load is equal to 85 percent of the breaking strength of the cable.

5.3.2.2 Non-Flexible Cable.- The sample splices shall be subjected to the Full Strength load specified in the above table. This load is equal to 95 percent of the breaking strength of the cable.

5.3.3 Joint Examination (Non-Flexible Cable Only).- After the Full Strength test, the serving wire shall be unwound to see that all parts are properly secured and the space between the cables and serving wire is well filled with solder.

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5.4 Evaluation of Results.-

5.4.1 Proof Load.- Terminals showing slipping, loosening, fraying, or any other signs of failure under the proof load shall be rejected.

5.4.2 Full Strength.- If failure occurs, it shall be ascertained whether the material or the workmanship is at fault. In case failure is attributed to the latter, the splicing process shall be immediately investigated and rectified.

5.4.3 Joint Examination (Non-Flexible Cable Only).- Evidence of slippage or loosening of the joint shall be cause for investigation of the methods used in preparing splices.

5.5 Acceptance Marking.- All accepted splice and cable assemblies shall be tagged with a permanent tag stamped with the official acceptance stamp, part number, date, and the Inspector's number. This tag shall be securely fastened to the cable in such a manner that it will not come loose.

6. NOTES

6.1 Proof loading, in addition to its use as a test, is an essential operation in the completion of a tight splice. The proving load takes out much of the slack in the splice, thus rendering the take-up of the turnbuckle more effective. As the prestretching of control cable assemblies just prior to installation and without subsequent coiling is required, the proof-loading test of the splice specified herein may be fulfilled by the prestretching of the spliced assembly.

6.2 Definitions.- The following definitions of splicing terms are applicable to this specification:

6.2.1 Standing Wire.- The measured length of cable which will be used when the splice is completed, is called the standing wire.

6.2.2 Bight.- The loop of cable passing around the thimble is called a bight.

6.2.3 Free End.- The loose end which is unstranded for splicing is called the free end.

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Air Force

Other interest:
Navy - BuAer

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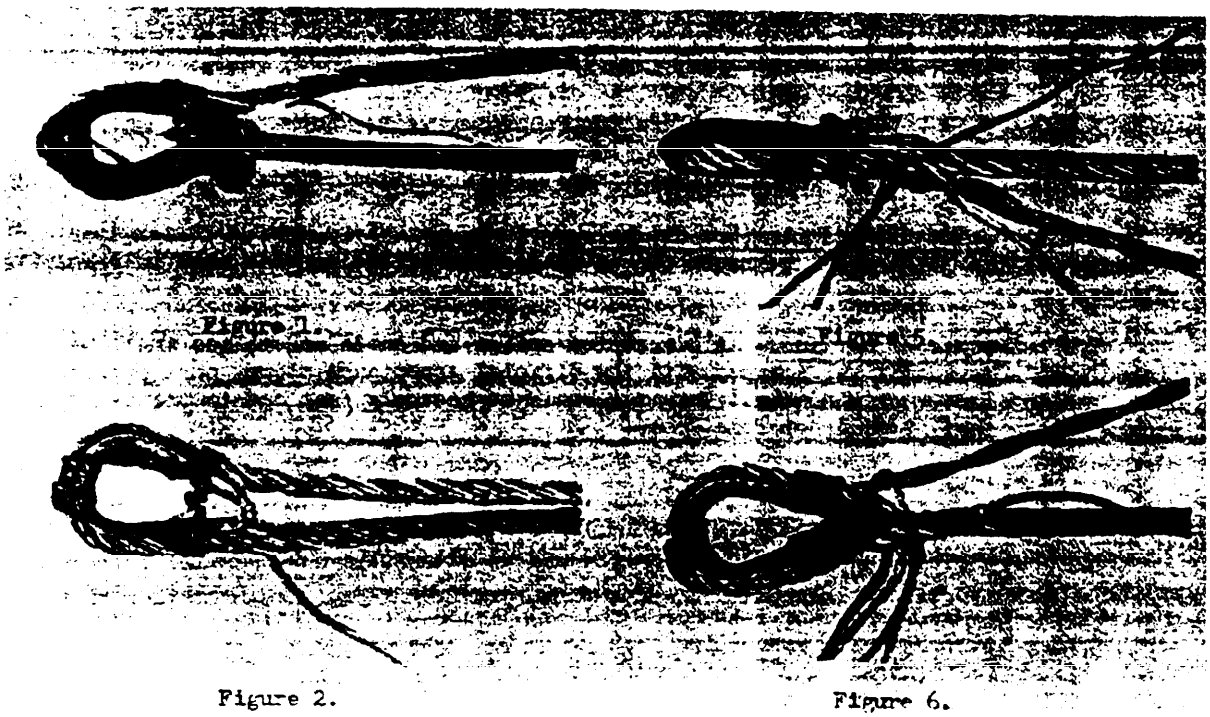


Figure 2.

Figure 6.

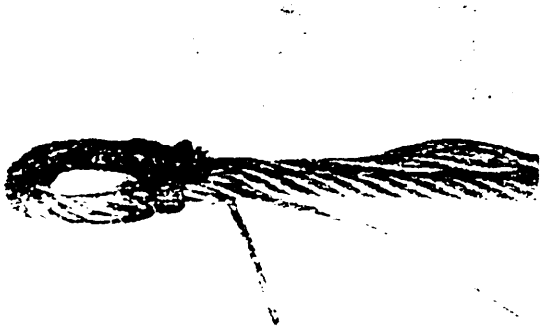


Figure 3.

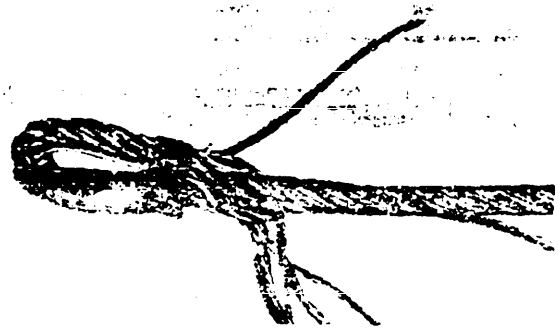


Figure 7.



Figure 4.

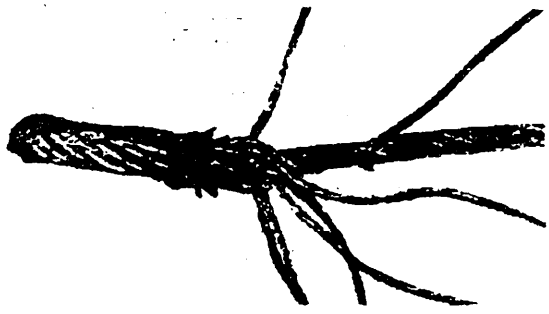


Figure 8.

Splicing Operations.

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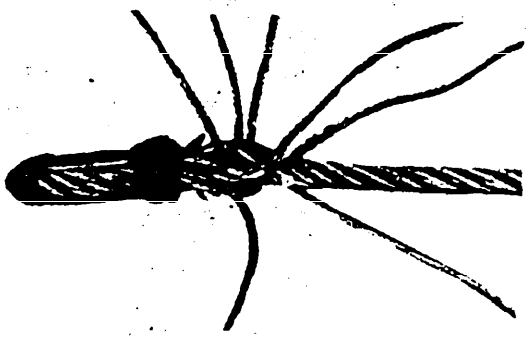


Figure 9.

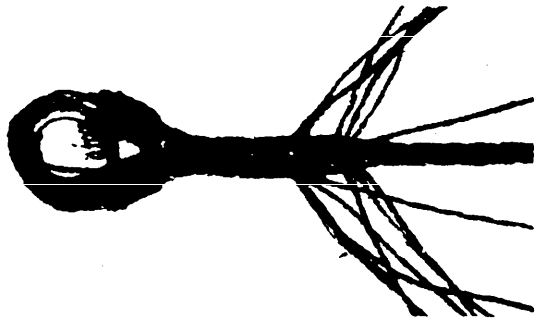


Figure 11.

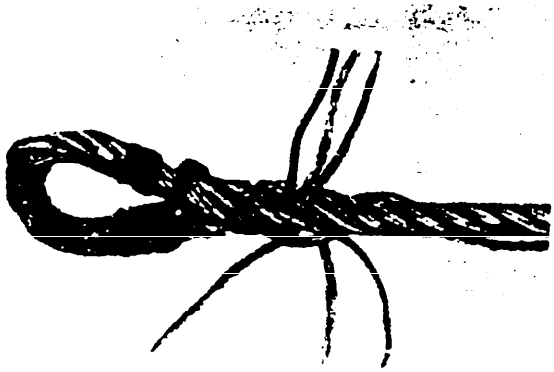


Figure 10.

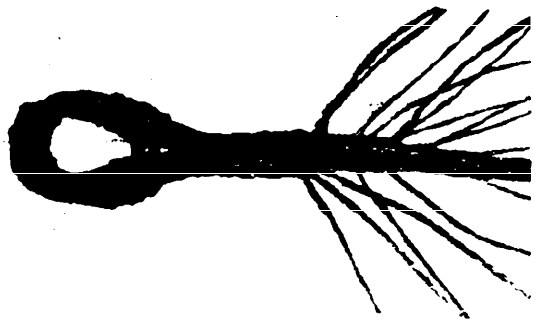


Figure 12.

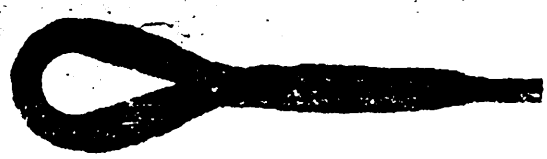


Figure 13.

Splicing Operations.

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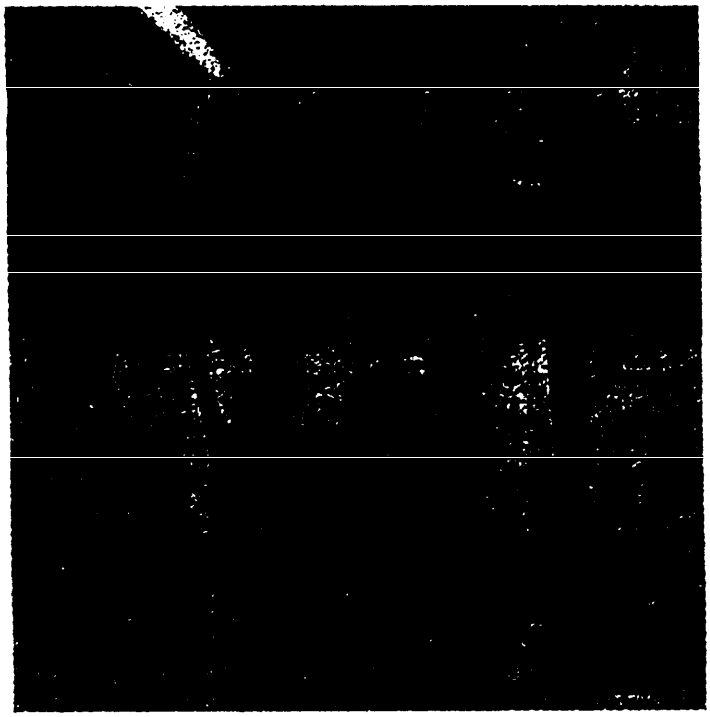


Figure 14. Tools for Cable Splicing.

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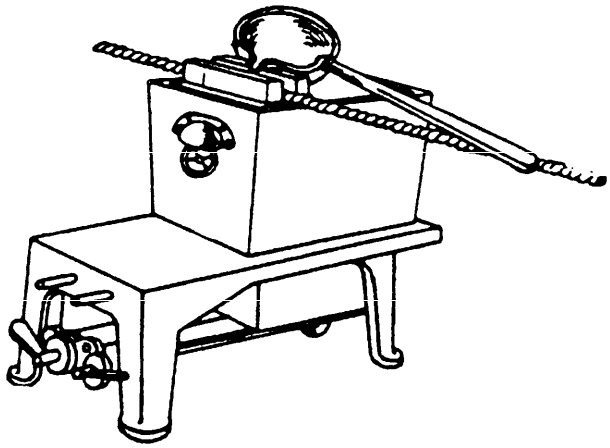
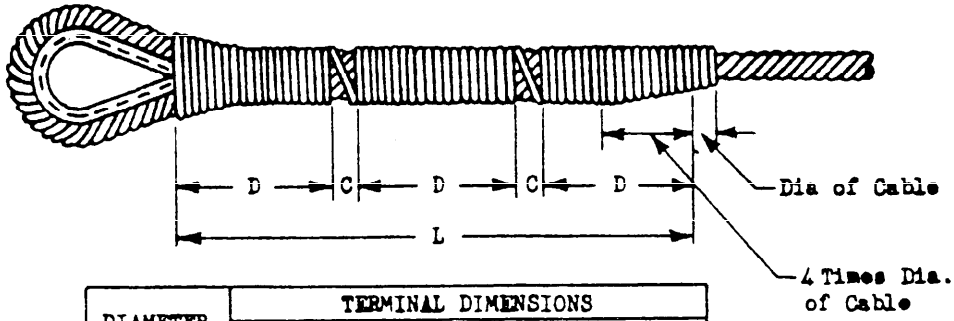


Figure 15. Soldering of Non-Flexible Cable Prior to Cutting.



DIAMETER OF CABLE	TERMINAL DIMENSIONS			
	L ±1/8	D ±1/16	C ±1/16	SERVING WIRE B & S GAUGE *
1/16	2	9/16	1/8	24
3/32	2-1/2	3/4	1/8	24
1/8	2-7/8	7/8	1/8	24
5/32	3-1/4	1	1/8	24
3/16	3-5/8	1-1/8	1/8	20
7/32	4	1-1/4	1/8	20
1/4	4-1/2	1-3/8	3/16	20
5/16	5-1/4	1-5/8	3/16	20
3/8	6-1/4	1-15/16	1/4	18
7/16	7	2-3/16	1/4	18
1/2	8	2-1/2	1/4	18

NOTE: *Tolerance ±2 Gauge Sizes.

Figure 16. Dimensions for Non-Flexible Cable Terminal Splice.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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PLATE NO. 15419 18A

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